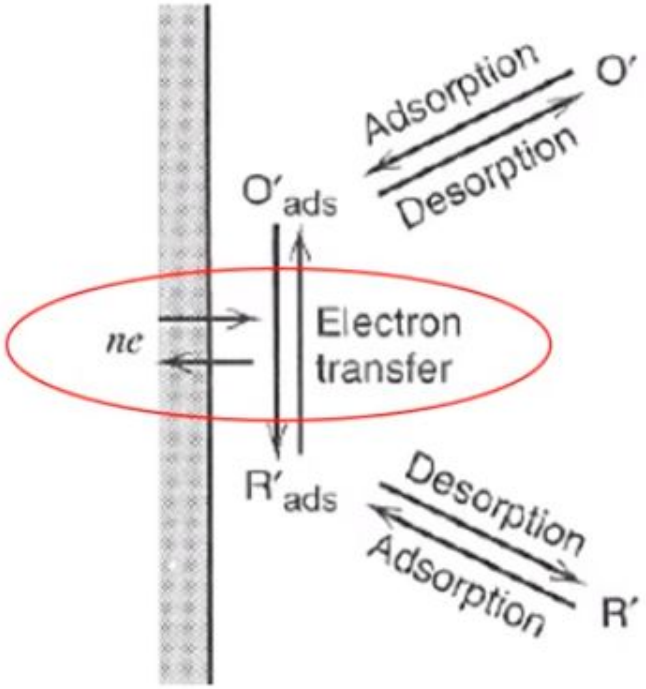
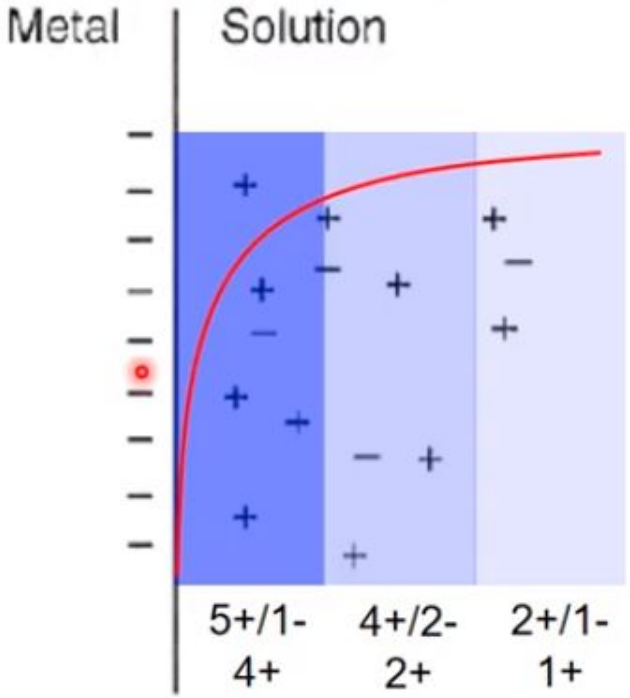
# Faradaic versus Non-faradaic

## Faradaic process or reaction



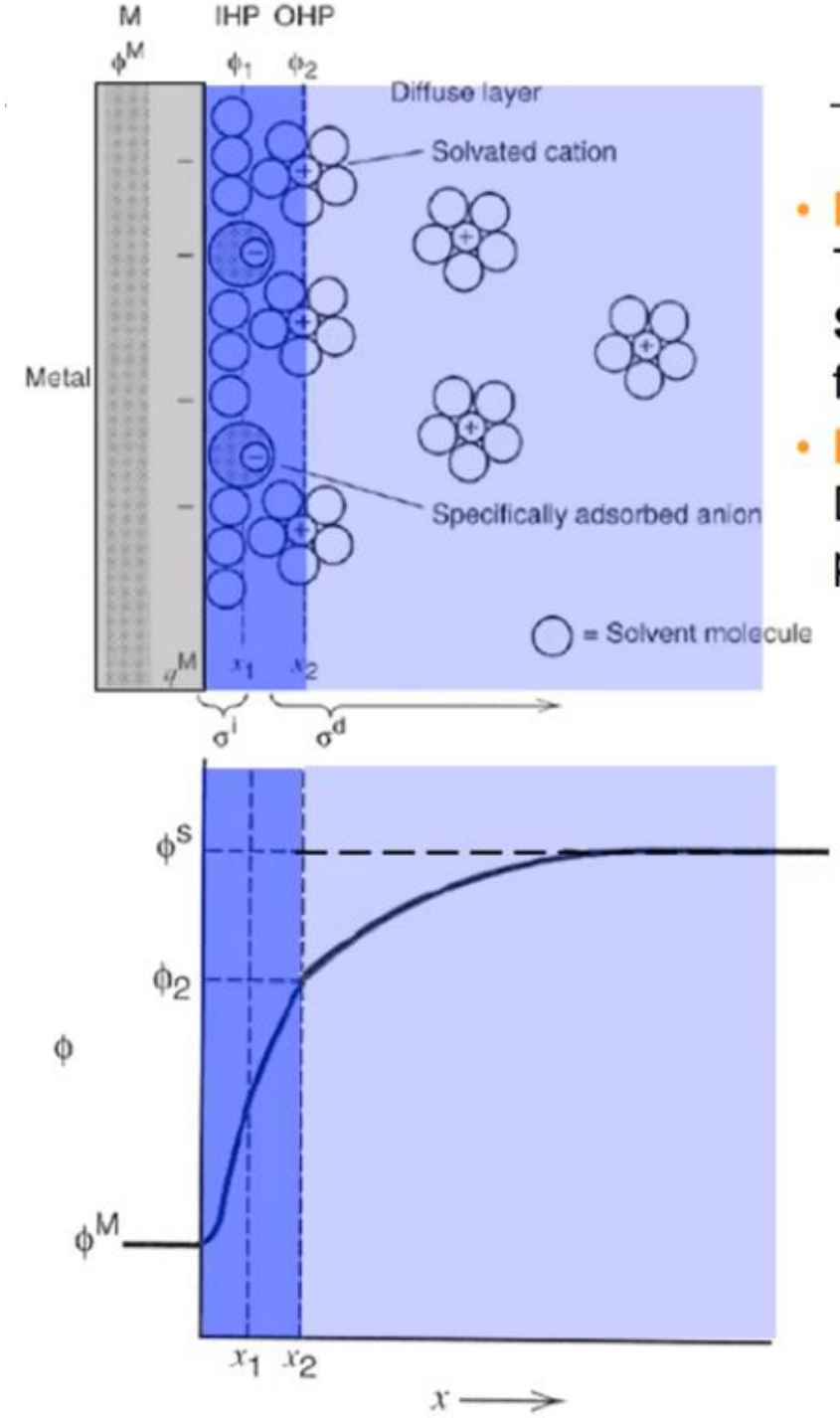

- Electron should pass the interface
- Driving chemical changes
- *Chemical reactions*

## Non-faradaic process

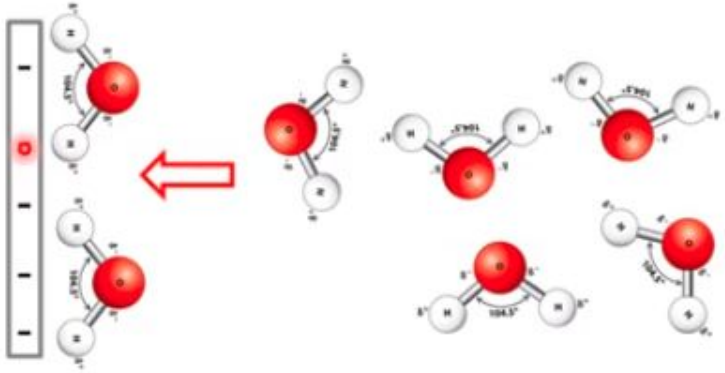



- Actually NOT lead to chemical changes
- Electrostatic forces btw different charges
- No electron can touch any species in electrolyte

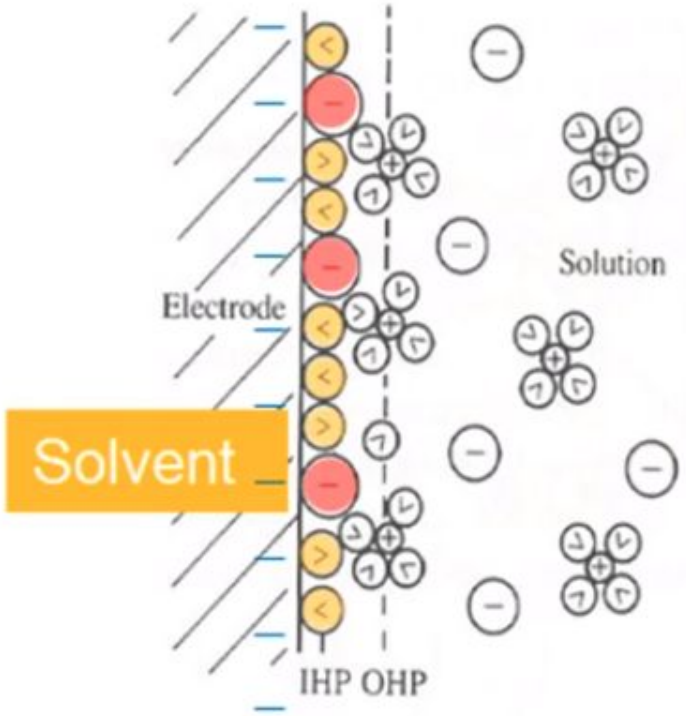
# Electric double layer (EDL)




- **Helmholtz layer**:
  Two sub-layers just on surface.
  **Solvent-molecular layer**
  followed by the layer of **solvated counter-ions.**
- **Diffuse layer**:
  Distributed counter-ions compensate the rest of potential difference.



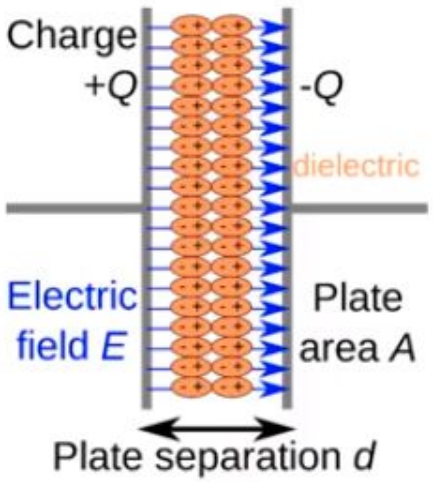
- EDLs are always developed even without applied potential.
- When a solid is immersed into a liquid, the reduction potential difference between the solid and the liquid drives the EDL formed.

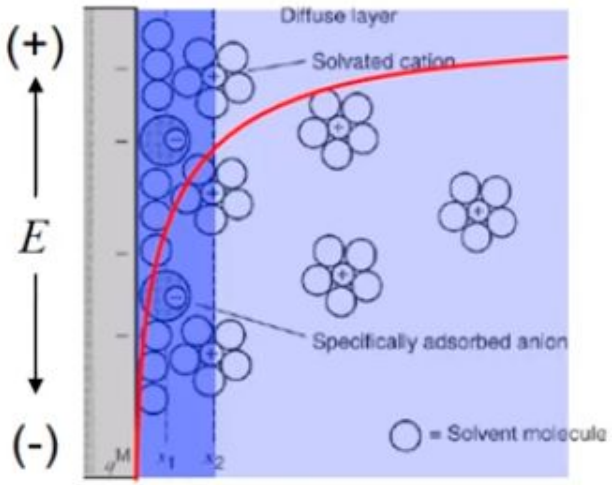



| | Polarity |
|---|---|
| Solvent | Weak (Dipole) |
| Cation | Strong (Attractive) |
| Anion | Strong (Repulsive) |
| Specifically adsorbed species | Very Strong |

# Electric double layer (EDL)

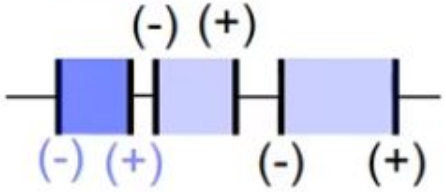
Electric double layers can be modeled by capacitors.




**Capacitor**

- A device that stores electrical energy in an electric field
- A dielectric material sandwiched between two metal plates
- One plate is positively charged while the other is negatively charged.
- Dipoles of the dielectric materials are oriented and re-arranged along the direction of applied electric field.

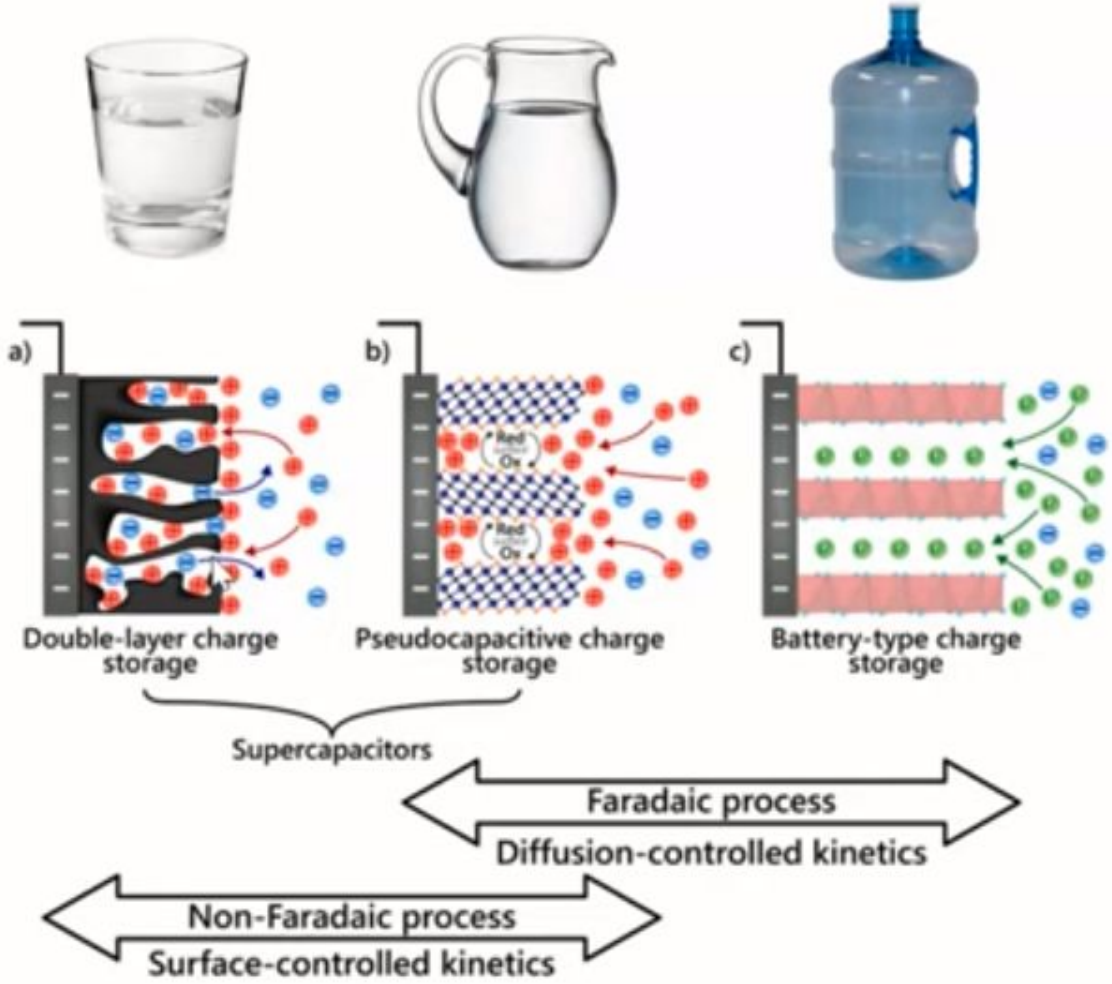



$C_H$ $\quad 1/C_d = 1/C_{d,1} + 1/C_{d,2} + \ldots$

Helmholtz capacitor

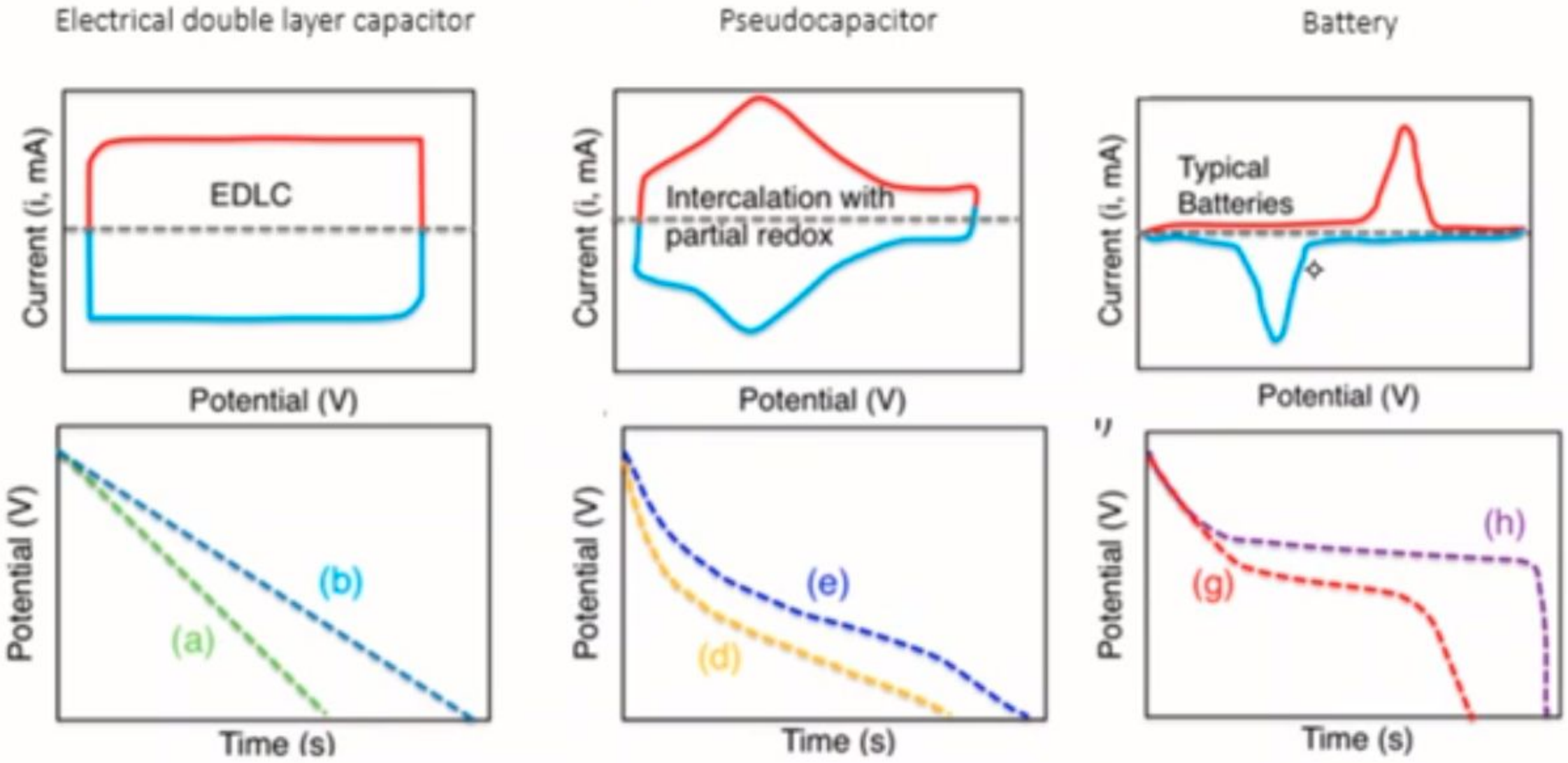
$$1/C_{dl} = 1/C_H + 1/C_d$$

Electric double layer capacitance




Mathis et al. Advanced Energy Materials, 2019, 9 (39): 1902007c

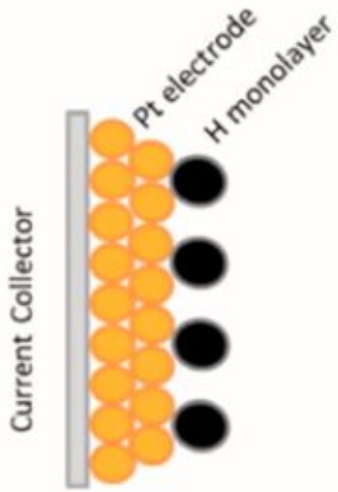
# Types of Energy Storage




*Advanced Energy Materials*, 2019, 9 (39): 1902007

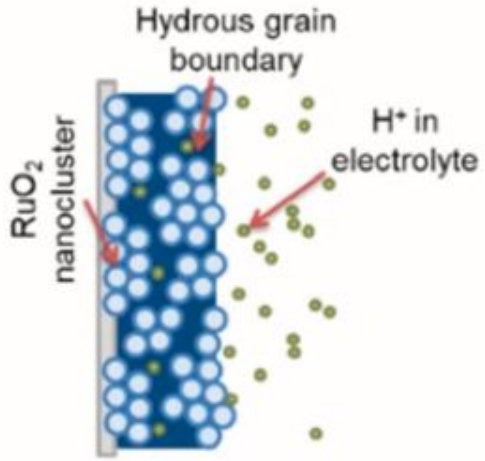
# Types of pseudocapacitance

**A** **Adsorption Pseudocapacitance**



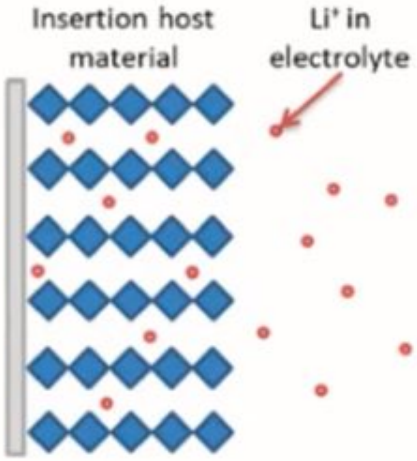

$$Pt + xH^+ + xe^- \leftrightarrow Pt{\cdot}xH_{ads}$$

**B** **Redox Pseudocapacitance**




$$RuO_2{\cdot}0.5H_2O + xH^+ + xe^- \leftrightarrow H_xRuO_2{\cdot}0.5H_2O$$

**C** **Intercalation Pseudocapacitance**




$$Nb_2O_5 + xLi^+ + xe^- \leftrightarrow Li_xNb_2O_5$$